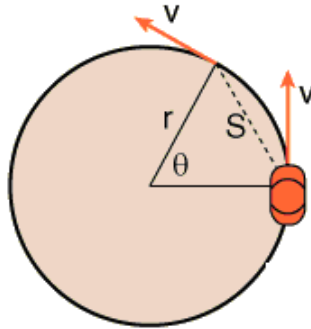


CIRCULAR MOTION

1. Consider an arc of a circle whose length (along the curve) is s , the circle is at radius r .



The angle measured, θ , measured in radians, is defined by the following equation :

$$\theta = \frac{s}{r}$$

The equation shows that the radian is a dimensionless quantity since it is the ratio of two lengths.

Definitions: **One radian** is that angle subtended by an arc length in a circle equal to the radius of the circle.

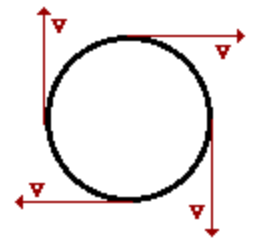
Rewriting the equation, $s = r\theta$

2. Differentiating with respect to time,

$$s = r\theta$$

$$\frac{\partial s}{\partial t} = r \frac{\partial \theta}{\partial t}$$

$$v = r\omega$$



The direction of the velocity vector at every instant is in a direction tangent to the circle.

Another way of looking into the above equation, when a body makes a complete circle:

- i) it has traveled a distance of $2\pi r$
- ii) the time taken is T , period

Therefore: $v = \frac{2\pi r}{T}$ or $v = 2\pi r f$

$v = r\omega$ where $\omega = 2\pi f$. angular speed
 2π = angular displacement

v – tangential speed

ω – angular speed

Define ω :

Rate of change of angular displacement.

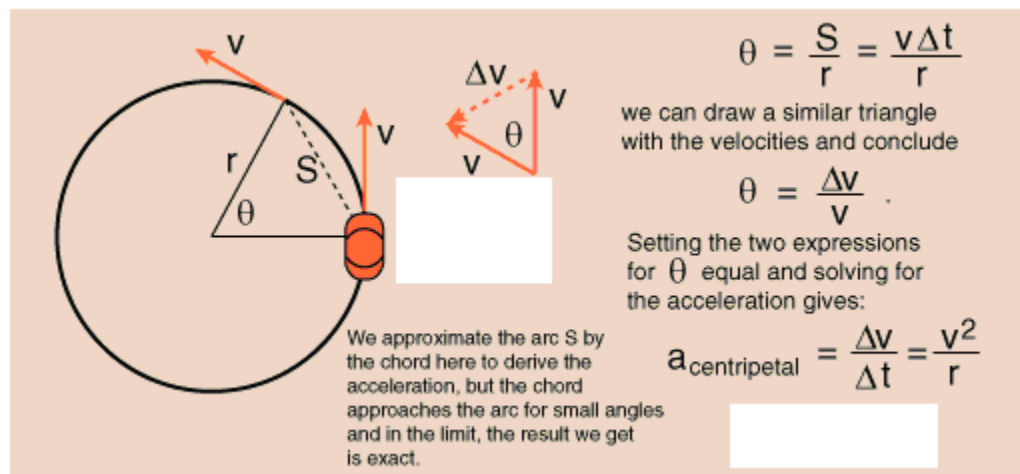
Ex:

1. Calculate the tangential speed and angular speed for a body on earth.
($R_E = 6400 \text{ km}$, $T = 24 \text{ hr}$)
2. A train is travelling on a track, which is part of a circle of radius 600 m , at a constant speed of 50 ms^{-1} . What is its angular velocity?
3. A washing machine spins its tub at a rate of $1200 \text{ revolutions per minute (rpm)}$. If the diameter of the tub is 35 cm , find
 - a) the angular velocity of the tub.
 - b) the linear speed of the rim of the tub.

.....

3. When a body is moving at constant speed, yet is said to be accelerating. Explain this statement:

-- An object moving in a circle at a constant speed, its direction is constantly changing. Based on NFL, the body experiences a constantly changing velocity. Therefore the body must be accelerating.



This acceleration is directed towards a fixed point - the center of the circle.

$$\text{Since } v = r\omega, \text{ then acceleration, } a_c = r\omega^2 = v\omega.$$

Even if moving around the perimeter of the circle with a constant speed, there is still a change in velocity and subsequently an acceleration. This acceleration is directed towards the center of the circle.

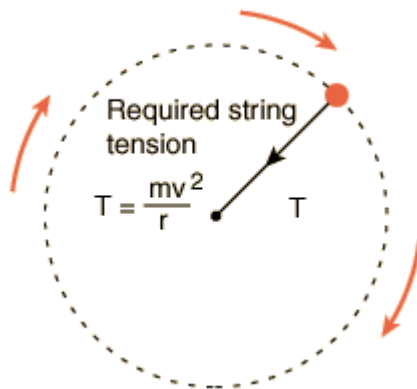
And in accord with Newton's second law of motion, an object which experiences an acceleration must also be experiencing a net force; and the direction of the net force is in the same direction as the acceleration.

$$\text{Therefore, } F_c = ma_c = mr\omega^2 = m v\omega.$$

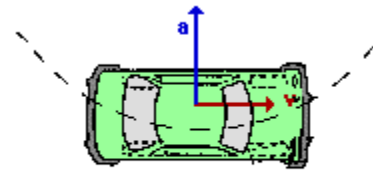
Question: Since there is a force acting on the body, explain whether there is a work done on the body?

1. Find the acceleration of the moon if it travels at a constant speed of 1020 ms^{-1} and takes 27.3 days for a complete revolution of the Earth.
2. A car is negotiating a circular track of radius 120 m at 75 kmh^{-1} , calculate its centripetal acceleration.
3. From question 5, say the mass of the driver is 70 kg and the car's mass is 1500 kg. Calculate the force acting on the driver and the car.

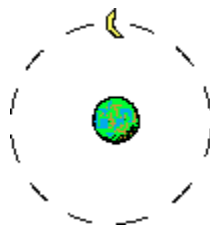
.....
4. Examples of Centripetal forces:



Car in Motion Makes a Left-Hand Turn



A passenger in motion would remain in motion with the same speed and in the same direction, thus causing the "sensation of an outwards acceleration."



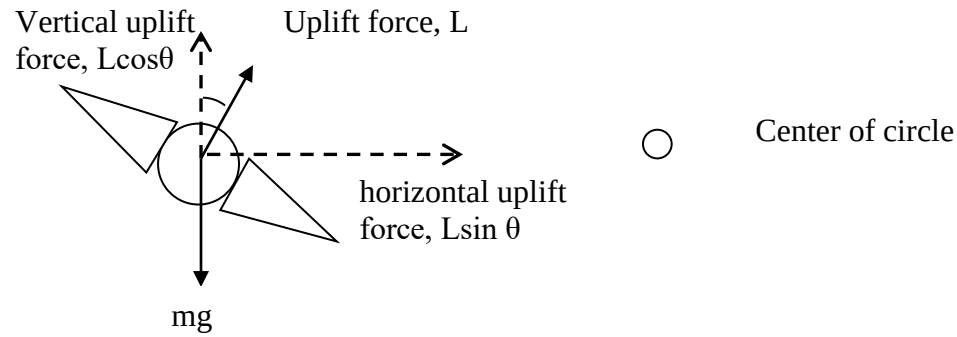
As the moon orbits the Earth, the force of gravity acting upon the moon provides the centripetal force required for circular motion.

If there is a centripetal force acting on the moon, explain why the moon does not fall down towards earth?



As a bucket of water is tied to a string and spun in a circle, the force of tension acting upon the bucket provides the centripetal force required for circular motion.

****Explain why water in the bucket does not spill out of the bucket?**



When a body moves in a circular path,

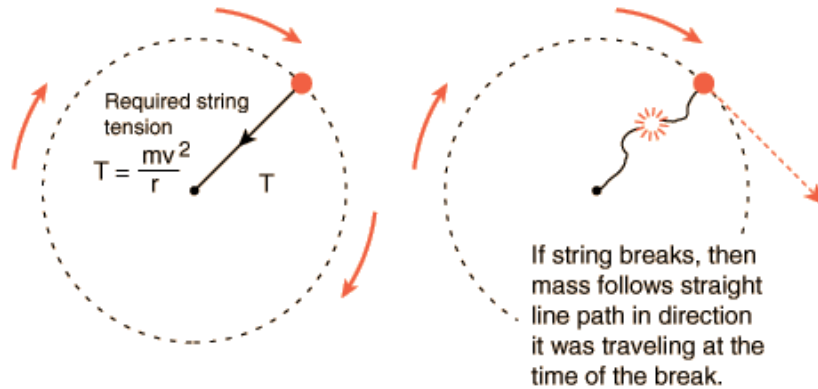
$$L\cos\theta = mg$$

Required centripetal force,
$$L\sin\theta = \frac{mv^2}{r}$$

$$\tan\theta = \frac{v^2}{rg}$$

- I. With constant radius, r the angle it should tilt increases with increasing speed of the body moving.
- II. By increasing the radius, a body can travel with a higher speed.

Question: In all the examples above, explain what will happen to rotating bodies if there is no centripetal force?



The string must provide the necessary centripetal force to move the ball in a circle. If the string breaks, the ball will move off in a **straight line with constant speed**.

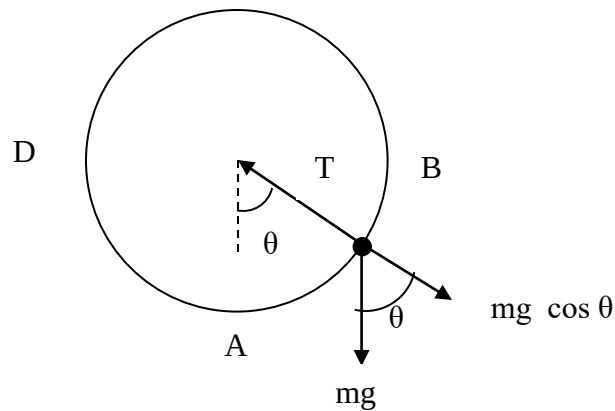
The straight line motion in the absence of the constraining force is an example of Newton's first law. The example here presumes that **no other net forces** are acting, such as horizontal motion on a frictionless surface.

If there is gravitational force then the body will have a parabolic path.

.....

4. An audio CD rotates 500 rpm when reading an inside track. The inside track of the CD is 50 mm out from the center. Find
 - i) the period of the rotation of the CD
 - ii) the linear speed and the angular speed
 - iii) the centripetal acceleration of the point on the inside track.
5. A satellite of mass 800 kg is orbiting the Moon in a circular path of radius 1760 km. The weight of the satellite is 1300 N. Find the speed of the satellite and its period of rotation.
6. A motorcycle stunt rider of mass 63 kg travels around the inside of a sphere of radius 3.0 m. What is the minimum speed he must have to be able to remain in contact with the sphere when he is upside down?
7. A coin will rest on a long playing record rotating at 45 rpm provided that it is not more than 10 cm from the centre of the record. How far away from the centre may it be placed if it is to remain on the record when rotated at 33.3 rpm.

5. Vertical circular motion C



At A, the angle is 0:

The required centripetal force, $F_c = \frac{mv^2}{r}$

$$T - mg \cos \theta = \frac{mv^2}{r}$$

$$T = \frac{mv^2}{r} + mg \cos \theta$$

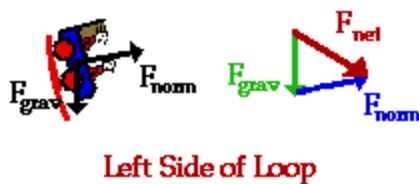
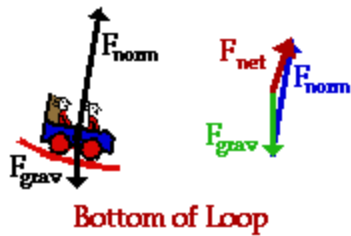
At A, the tension is maximum.

At B and D, the angle is 90° and 270° , $\cos \theta = 0$ Therefore, $T = \frac{mv^2}{r}$

At C, $\theta = 180^\circ$, $\cos 180^\circ = -1$

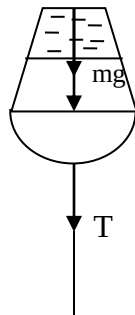
$$\text{Therefore, } T = \frac{mv^2}{r} - mg \cos \theta$$

At C, the tension is minimum.



Feelings of weightlessness and heaviness are associated with the normal force; they have little to do with the force of gravity. A person who feels weightless has not lost weight. The force of gravity acting upon the person is the same magnitude as it always is. Observe that in the diagram above the **force of gravity is everywhere the same**. The **normal force** however has a **small magnitude at the top of the loop** (where the rider often feels weightless) and a **large magnitude at the bottom of the loop** (where the rider often feels heavy).

The normal force is large at the bottom of the loop because in order for the net force to be directed inward, the normal force must be greater than the **outward** gravity force. At the top of the loop, the gravity force is directed inward and thus, there is no need for a large normal force in order to sustain the *circular* motion. The fact that a rider experiences a large force exerted by the seat upon her body when at the bottom of the loop is the explanation of why she *feels* heavy. In actuality, she is not heavier; she is only experiencing the large magnitude of force which is normally exerted by seats upon heavy people while at rest.



If the water with a mass of m in the bucket not to spill:

$$R > 0$$

$$\frac{mv^2}{r} - mg \cos\theta > 0$$

$$\frac{mv^2}{r} > mg$$

$$v^2 > rg$$

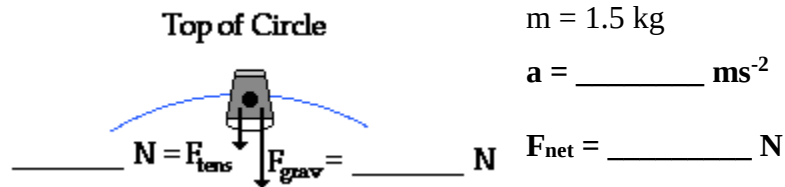
$$v > \sqrt{rg}$$

Minimum speed at the top $v = \sqrt{rg}$

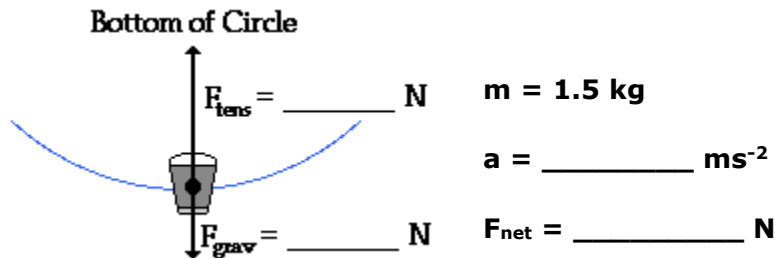
What will be the minimum speed of the water at bottom, if the water should not spill at the top?

Hint : conservation of energy.

1. A 1.5-kg bucket of water is tied by a rope and whirled in a circle with a radius of 1.0 m. At the top of the circular loop, the speed of the bucket is 4.0 ms^{-1} . Determine the acceleration, the net force and the individual force values when the bucket is at the top of the circular loop.



2. A 1.5-kg bucket of water is tied by a rope and whirled in a circle with a radius of 1.0 m. At the bottom of the circular loop, the speed of the bucket is 6.0 m/s . Determine the acceleration, the net force and the individual force values when the bucket is at the bottom of the circular loop.



1. A bucket, containing 1.5 kg water, is tied by a rope and whirled in a circle of radius 1.0 m as shown in **Fig. 2**.

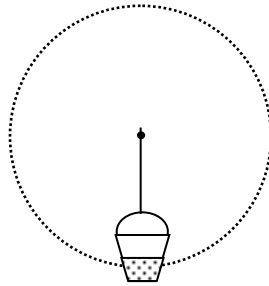


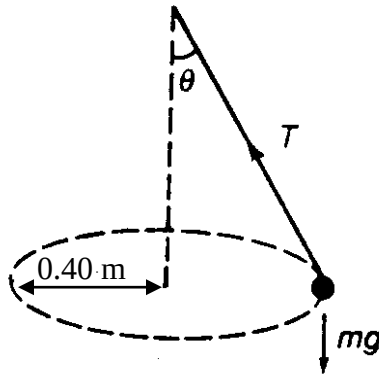
Fig. 2

i. At the bottom of the circular loop, the speed of the bucket is 4.0 ms^{-1} . Determine the centripetal force acting on the water.

ii. On **Fig. 2**, mark, with labelled arrows, the direction of the weight of water W and the reaction force R acted by the bottom of bucket on the water.

iii. Calculate the reaction force R .

2. A conical pendulum consists of a small bob of mass 0.20 kg attached to an inextensible string of length 0.80 m . The bob rotates in a horizontal circle of radius 0.40 m , of which the centre is vertically below the point of suspension. (Assume $g = 10 \text{ ms}^{-2}$)



Write down an expression that relates the string tension, T with the linear speed, v of the bob.

..... [2]

- a) Write down an expression that shows equilibrium of forces in the vertical direction.

.....[1]

- b) Calculate the linear speed of the bob in ms^{-1} .

linear speed = [3]

- c) Calculate the period of rotation of the bob.

period = [2]

- d) Find the tension in the string.

tension = [2]

- 3 (a) Explain what meant by angular velocity.

.....
[1]

- (b) An object travelling at a constant speed in a circular motion, as shown in the Fig.1 below, is said to have a centripetal acceleration. Using the diagram, explain

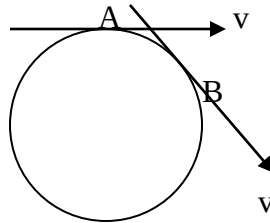


Fig. 1

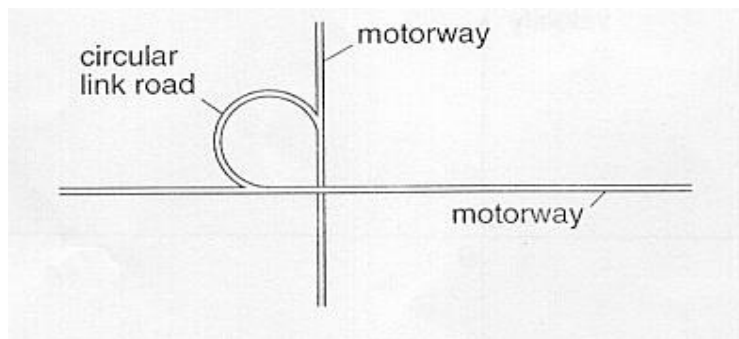
- (i) why there is an acceleration even though the speed is constant.

.....
[2]

- (ii) the direction of the acceleration.

.....[1]

- (c) A motorway designer plans to have motorists leaving one motorway and joining another by constructing a circular link road, as shown in Fig.2



In order to use as small an area of land as possible, the designer proposed a speed limit of 25 ms^{-1} for cars on the circular link road.

- (i) Calculate the minimum radius for the circular link road, given that the maximum sideways force between a car and the road is $0.8 W$, where W is the weight of the car.

[3]

- (ii) Suggest why buses, which are taller than cars, have to go at a slower speed than the 25 ms^{-1} speed for the cars.

.....
.....
.....[2]

4. The Moon travels round the Earth in a circular orbit of radius $3.8 \times 10^8 \text{ m}$ with a period of 27.3 days. The mass of the Moon is $7.4 \times 10^{22} \text{ kg}$.

- i. Calculate the Moon's angular velocity.
- ii. Determine the force required to keep the Moon moving in its orbit.
- iii. Name the force existing between the Moon and the Earth that provides the centripetal force. Hence, determine the mass of the Earth.

[9]

5. (a) A car goes round a curve in a road at constant speed. Explain why, although its speed is constant, it is accelerating.

.....

.....

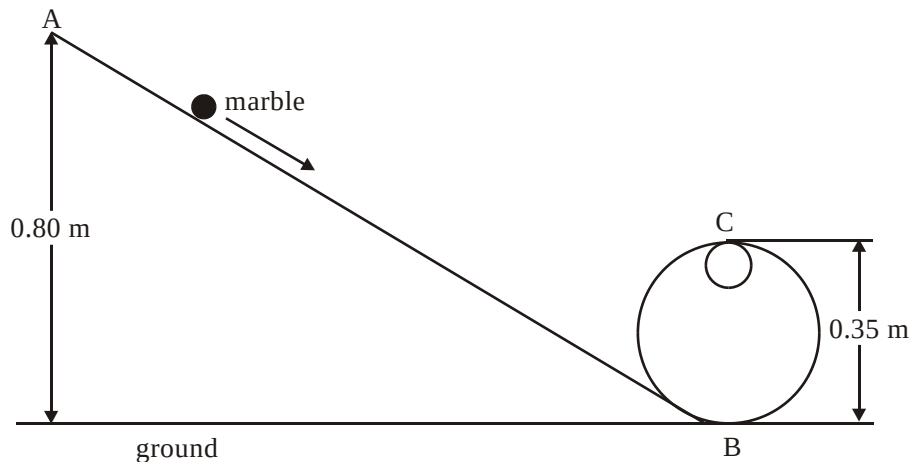
.....

.....

.....

(2)

In the diagram below, a marble (small glass sphere) rolls down a track, the bottom part of which has been bent into a loop. The end A of the track, from which the marble is released, is at a height of 0.80 m above the ground. Point B is the lowest point and point C the highest point of the loop. The diameter of the loop is 0.35 m.



The mass of the marble is 0.050 kg. Friction forces and any gain in kinetic energy due to the rotating of the marble can be ignored.

Consider the marble when it is at point C.

- (b) (i) On the diagram opposite, draw an arrow to show the direction of the resultant force acting on the marble.

(1)

- (ii) State the names of the **two** forces acting on the marble.

.....

.....

(2)

(iii) Deduce that the speed of the marble is 3.0 ms^{-1} .

.....

.....

.....

.....

(3)

(iv) Determine the resultant force acting on the marble and hence determine the reaction force of the track on the marble.

.....

.....

.....

.....

.....

(4)

(Total 12 marks)

<http://www.mcasco.com/p1acf.html>

http://www.phy.ntnu.edu.tw/oldjava/circularMotion/circular3D_e.html

<http://www.physclips.unsw.edu.au/jw/circular.htm>